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DATA STRUCTURES AND ALGORITHMS – 3

Pathogen Genomic Surveillance & Outbreak-Tracking System

Team 6 | Section 13

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GUIDE

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Abstract

The Pathogen Genomic Surveillance & Outbreak-Tracing System is a data-driven system designed to monitor infectious pathogens, identify potential disease outbreaks, and support investigation of pathogen transmission.

It combines genomic sequence data with epidemiological information — patient location, sample collection date, pathogen type, and case details — so that genetic similarities and differences among pathogens can be used to identify related cases and possible outbreak clusters.

Bioinformatics techniques process and compare genomic sequences to find closely related samples, organizing cases by geography and time to reveal how an outbreak may be developing and spreading. Cluster diagrams, geographical maps, and outbreak timelines present these findings clearly for researchers, laboratories, and public-health authorities.

Key Highlights

- Genomic sequence comparison via alignment algorithms
- Automated outbreak cluster identification
- Geographic + temporal case organization
- Visual maps, timelines & cluster diagrams

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Introduction

Infectious diseases can spread rapidly, making it difficult to identify related cases, find possible sources of an outbreak, and understand how the disease is spreading. Traditional disease surveillance mainly depends on clinical and epidemiological information such as patient details, location, and sample collection dates. Genomic data provides additional information by showing how closely different pathogen samples are related.

The system combines pathogen genomic sequences with patient, location, and time-based information to identify genetically related cases and possible outbreak clusters — applying String Algorithms and Dynamic Programming from DSA-3.

Core Techniques Applied

Needleman-Wunsch

Global sequence alignment

Smith-Waterman

Local sequence alignment

Wagner-Fischer

Edit-distance similarity scoring

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Literature Survey

Traditional Epidemiological Surveillance

Relies on patient records, location, and symptom-based case reporting. Effective for broad monitoring but cannot confirm genetic relatedness between cases.

Sequence-Alignment Tools (e.g. BLAST)

Widely used for comparing biological sequences and finding homologous regions, but generally used as a standalone search tool rather than integrated with outbreak-cluster logic.

Genomic Epidemiology Platforms

Tools such as Nextstrain and GISAID enable pathogen phylogenetics and global data-sharing, but are oriented toward large-scale visualization rather than case-level cluster investigation.

Gap Identified

Existing approaches largely treat genomic analysis, geography, and time as separate signals — this project integrates all three into one alignment-driven clustering and visualization workflow.

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Problem Statement and Objectives

Problem Statement

Clinical and location data alone cannot confirm whether two cases are genetically related. Genomic, geographic, and time-based evidence are typically analyzed in isolation, which slows manual confirmation of outbreak clusters and delays public-health response.

Objectives

- Compare pathogen genomes using alignment & edit-distance algorithms
- Detect outbreak clusters automatically from related cases
- Organize cases by geography to trace spread across regions
- Build outbreak timelines from sample collection dates
- Visualize findings clearly for researchers & public-health teams

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Hardware and Software Requirements

Hardware

- **OS:** Windows 10/11 or Ubuntu
- **Processor:** Intel i5 / Ryzen 5 or higher
- **RAM:** 8 GB minimum
- **Storage:** 256 GB SSD or higher
- **Network:** Stable internet connection

Software

- **Language:** Python 3.x
- **Libraries:** Pandas, NumPy, Biopython, Scikit-learn
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Flask / FastAPI
- **Database:** MySQL / PostgreSQL
- **Tools:** Jupyter / VS Code, Git

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Results

Expected system output at project completion

Cluster Diagrams

Genetically related samples grouped visually by similarity score.

Geographic Maps

Case locations plotted spatially to reveal spread across regions.

Outbreak Timelines

Sample dates arranged chronologically to track cluster growth.

07 Conclusion

By combining genomic sequence alignment, epidemiological context, and clear visualization, the system helps identify outbreak clusters faster and gives public-health teams evidence-based grounds for disease-control decisions.

- Higher-resolution outbreak detection than clinical data alone
- Faster identification of related cases across regions
- Actionable output for researchers & public-health authorities

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Future Scope

Real-time data feeds

Integrate live sequencing and case-report feeds for continuous surveillance.

ML-based cluster prediction

Apply machine learning to flag likely emerging clusters earlier.

Public-health integration

Connect with regional/national health databases for wider adoption.

Multi-pathogen support

Extend beyond a single pathogen family to broader surveillance use.